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OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 8: Sample and Hold

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8 Sample and Hold

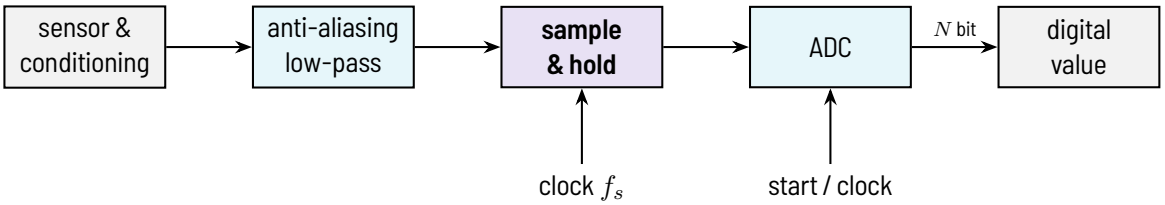
8.1 Why Sample and Hold?

The previous chapters produced a clean, filtered analog voltage. The next step is to turn it into numbers. A digital system cannot store a continuous signal; it takes one value per sampling period and works with this sequence of numbers. The circuit that takes these values is the *sample and hold* (S&H) stage. It sits between the anti-aliasing filter and the analog-to-digital converter (ADC).

The S&H stage has two jobs. First, it defines the *sampling instant*: the moment at which the signal is measured. Second, it keeps the measured voltage constant while the converter works on it. Many converters need some time for a conversion and give wrong results if their input changes during this time. A successive-approximation converter, for example, compares the input with a sequence of trial voltages; if the input moves in between, the result is a mixture of different input values.

In the lab and in the capstone task the sampling frequency is very low, $f_s = 5$ Hz, so that we can follow every sample by eye on the Falstad scope and on the LEDs. The principles are the same as in a converter with millions of samples per second, only the time scales differ [6, 7].

Position in the signal chain



- the S&H stage **defines the sampling instant** $t_k = k\,T_s$, $T_s = 1/f_s$
- it keeps the voltage **constant while the ADC converts**
- capstone: $f_s = 5$ Hz, 3-bit flash converter, full scale 4 V

What goes wrong without S&H

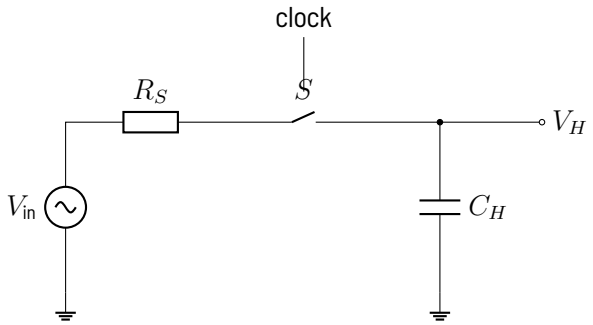
- the input changes during the conversion time t_{conv}
- the error must stay below $\frac{1}{2}$ LSB: $\left|\frac{dV}{dt}\right|_{\max} t_{\text{conv}} < \frac{1}{2}\,\text{LSB}$
- sine $V(t) = \hat{V}\sin(2\pi f t)$: maximum slope $2\pi f \hat{V}$
- example: 12-bit ADC, 4 V full scale (LSB $= 0.98$ mV), $t_{\text{conv}} = 10$ µs, $\hat{V} = 2$ V
  - $f < \frac{\text{LSB}/2}{2\pi \hat{V} t_{\text{conv}}} = 3.9$ Hz without S&H
  - with S&H only the much shorter *aperture time* counts (nanoseconds)
- a **flash** converter decides in one step, but the digital outputs still have to be read and displayed at a defined instant

The example shows how strict the requirement is. A 12-bit converter without S&H can only follow signals of a few hertz, even though its conversion time is short. With an S&H stage in front, the relevant time is the uncertainty of the sampling instant, which is many orders of magnitude shorter. In our capstone the flash converter is fast, but the S&H stage still gives us one well-defined value per 200 ms, so that the display does not flicker and the digital code belongs to a known instant.

8.2 The Track-and-Hold Principle

The simplest S&H stage is a switch and a capacitor. While the switch is closed, the capacitor voltage follows the input (track mode). When the switch opens, the capacitor keeps its charge and therefore its voltage (hold mode). Strictly speaking this circuit is a *track-and-hold* circuit, because it follows the input during the whole time the switch is closed. A true sample-and-hold would take a very short sample. In practice the two names are used for the same circuit.

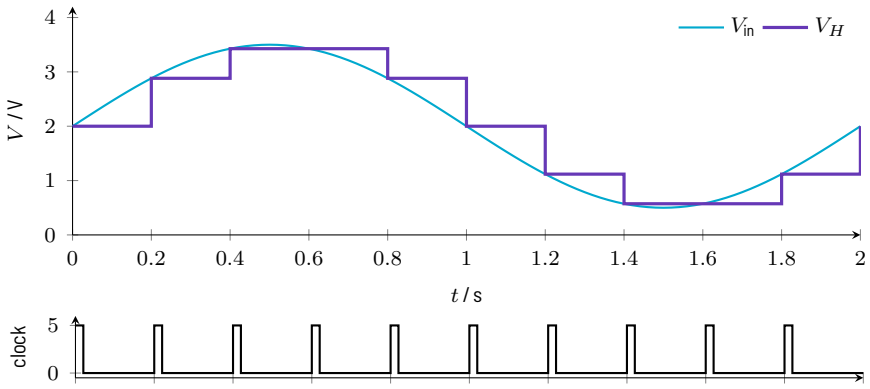
Switch and hold capacitor



- clock high: switch closed, C_H follows V_{in} (**track**)
- clock low: switch open, C_H keeps its charge (**hold**)
- capstone clock: 5 Hz, duty cycle $10\,\%$: track 20 ms, hold 180 ms

R_S is the total resistance in the charging path. It contains the output resistance of the source, the on-resistance of the switch and, in our circuits, a resistor of $1\,\text{k}\Omega$ that limits the charging current of the op-amp that drives the capacitor.

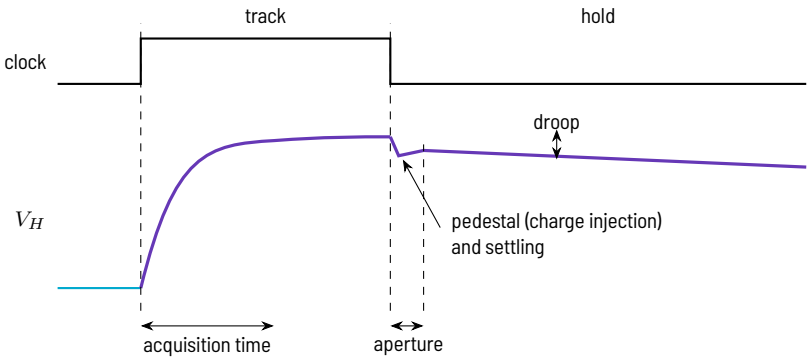
Timing diagram



- the output is a **staircase**: one step per sampling period
- the value of each step is V_{in} at the **end** of the track phase

The plot shows $V_{\text{in}} = 2\,\text{V} + 1.5\,\text{V}\,\sin(2\pi \cdot 0.5\,\text{Hz} \cdot t)$ sampled at $5\,\text{Hz}$. The clock in the timing row is high for one tenth of each period. The steps of V_H start at the falling clock edge, when the switch opens. (In the plot the short track phases are not resolved.)

Terms



- **acquisition time**: track time needed until V_H is within the error band (e.g. $\frac{1}{2}$ LSB)
- **aperture time**: delay between the hold command and the real opening of the switch; its variation is the **aperture jitter**
- **pedestal**: step at the transition to hold; **droop**: slow change during hold

Falstad: basic S&H

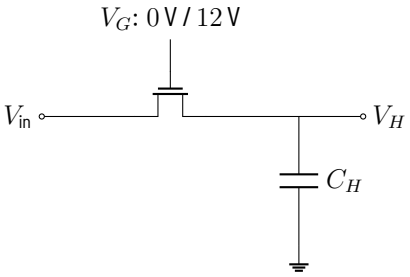
- $V_{\text{in}} = 2\,\text{V} + 1.5\,\text{V}\,\sin(2\pi \cdot 0.5\,\text{Hz} \cdot t)$, $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$
- clock rail: square wave 5 Hz, 0 V … 5 V, duty cycle 0.1
- held values: 2.094 V, 2.956 V, 3.452 V, 3.394 V, 2.804 V
- the scope shows V_{in}, V_H (label `Vhold`) and the clock

▶ Falstad: `examples/samplehold/sh_basic.txt` The held values are the input voltages at the end of each track window, $t = 0.02\,\text{s}, 0.22\,\text{s}, \ldots$ The simulation shows them about 5 mV lower, because the capacitor lags behind a rising input by roughly the slope times the time constant ($4.7\,\text{V/s} \cdot 1$ ms).

8.3 Analog Switches

The switch of an S&H stage is electronic. It must close and open on a logic signal, have a low resistance when closed and an extremely high resistance when open, and it must not disturb the stored charge. The standard solution is a MOSFET, or two complementary MOSFETs in parallel (transmission gate) [3, 1].

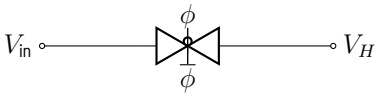
The MOSFET as a switch

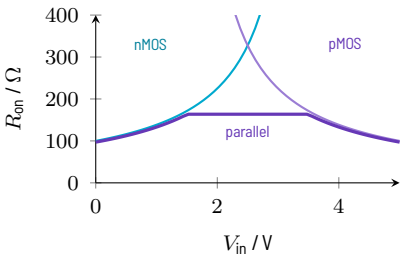


- n-channel MOSFET: conducts if $V_{GS} = V_G - V_{\text{in}} > V_{th}$
- on-resistance $R_{\text{on}} \approx \frac{1}{k\,(V_{GS} - V_{th})}$: depends on V_{in}!
- near $V_{\text{in}} \approx V_G - V_{th}$ the switch does not close any more
- off: leakage currents in the pA ...nA range

Source and drain of a MOSFET switch are interchangeable; the current can flow in both directions. The gate voltage must be higher than the largest input voltage plus the threshold voltage. If the input rises, the gate-source voltage shrinks and the on-resistance grows. A single n-channel switch is therefore good for signals near ground, a single p-channel switch for signals near the positive supply.

CMOS transmission gate





- nMOS and pMOS in parallel, driven with complementary signals
- nMOS conducts well near 0 V, pMOS near V_{DD}: together almost constant R_{on}
- integrated as CD4066, 74HC4066 (4 switches, $R_{\text{on}} \approx 50\,\Omega$ ... $300\,\Omega$), DG4xx

The analog switch in Falstad

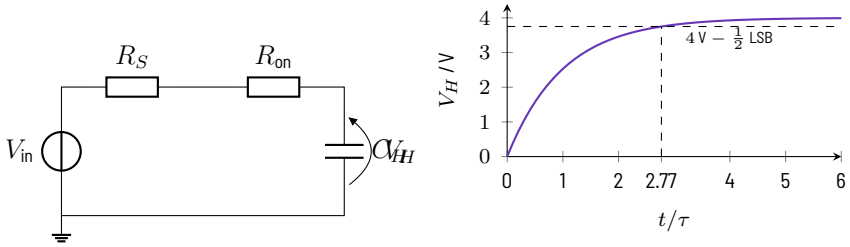
- element *Analog Switch* (models a FET switch / transmission gate)
  - closed if the control input is above 2.5 V
  - $R_{\text{on}} = 20\,\Omega$, $R_{\text{off}} = 10\,\text{G}\Omega$ (defaults, edit with a double click)
- no charge injection, no leakage: add them as extra elements if you want to see their effect
- in our circuits: $R_S = 1\,\text{k}\Omega$ in series
  - limits the current from the op-amp output into the empty capacitor
  - isolates the op-amp from the large capacitive load (stability)
  - real lab switches have R_{on} of some $100\,\Omega$ anyway

The Falstad switch is ideal apart from its two resistances. A real switch adds charge injection, leakage and a small coupling capacitance across the open switch. We add these effects one at a time with discrete elements (a capacitor from the clock to the hold node, a current source in parallel to C_H) to see their influence separately.

8.4 Designing the Hold Capacitor

The hold capacitor is the central design decision. Two requirements pull in opposite directions. A small capacitor charges quickly in the track phase, but it loses its voltage quickly through leakage currents in the hold phase. A large capacitor holds its voltage well, but it may not be charged completely in the short track window. Both requirements give a limit for C_H; any value in between works, and we choose one with margin on both sides.

Acquisition: charging in the track window



- $V_H(t) = V_{in}\,(1 - e^{-t/\tau})$ with $\tau = (R_S + R_{on})\,C_H$
- error after time t: $V_{in}\,e^{-t/\tau}$; worst case: step over the full scale V_{FS}

Settling to $\frac{1}{2}$ LSB

- requirement: $V_{FS}\,e^{-t/\tau} \le \frac{1}{2}\,\text{LSB} = \frac{V_{FS}}{2^{N+1}}$

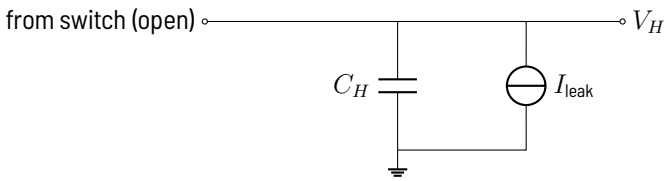
Acquisition time for N bit

$$t_{acq} = \tau\,\ln\left(2^{N+1}\right) = (N+1)\,\ln 2 \cdot \tau \approx 0.69\,(N+1)\,\tau$$

- $N = 3$: $t_{acq} = 2.77\,\tau$; $N = 8$: $6.2\,\tau$; $N = 12$: $9.0\,\tau$; $N = 16$: $11.8\,\tau$
- design rule: $t_{acq} \le t_{track} \quad\Rightarrow\quad C_H \le \dfrac{t_{track}}{(R_S + R_{on})\,\ln 2^{N+1}}$
- example: $R_S + R_{on} = 1.02\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$: $\tau = 1.02\,\text{ms}$, $t_{acq} = 2.83\,\text{ms} \ll 20\,\text{ms}$

▶ Falstad: `examples/samplehold/sh_acquisition.txt` The Falstad example charges $C_H = 1\,\mu\text{F}$ from a $4\,\text{V}$ source through $1\,\text{k}\Omega$ plus the $20\,\Omega$ of the switch. After one time constant ($t = 1.02\,\text{ms}$) the capacitor shows $2.53\,\text{V}$ ($63\,\%$), after $t_{acq} = 2.83\,\text{ms}$ it reaches $3.75\,\text{V}$, i.e. $\frac{1}{2}$ LSB below the final value, and at the end of the $20\,\text{ms}$ window it holds $4.00\,\text{V}$. Use a time constant of a few percent of the track window; the remaining time is reserve for the settling of the op-amp and for tolerances.

Droop: losing charge in the hold phase



- leakage currents: switch (off), capacitor dielectric, input bias current of the buffer, board surface

Droop

$$\frac{dV_H}{dt} = \frac{I_{leak}}{C_H} \quad\Rightarrow\quad \Delta V_{droop} = \frac{I_{leak}\,t_{hold}}{C_H} \quad\Rightarrow\quad C_H \ge \frac{I_{leak}\,t_{hold}}{\Delta V_{max}}$$

- t_{hold}: until the conversion is finished; worst case the whole period $T_s - t_{track}$

Too small, right, too large

- Falstad: $V_{in} = 3\,\text{V}$, $R_S = 1\,\text{k}\Omega$, leakage $I_{leak} = 50\,\text{nA}$ (exaggerated)

	$C_H = 10\,\text{nF}$	$C_H = 1\,\mu\text{F}$	$C_H = 100\,\mu\text{F}$
τ	$10\,\mu\text{s}$	$1.02\,\text{ms}$	$102\,\text{ms}$
charged in $20\,\text{ms}$?	yes	yes	only $18\,\%$ per sample
droop in $180\,\text{ms}$	$0.9\,\text{V}$	$9\,\text{mV}$	$0.09\,\text{mV}$
Falstad	$3.00\,\text{V} \to 2.10\,\text{V}$	$3.00\,\text{V} \to 2.99\,\text{V}$	$0.534\,\text{V}$, ..., $1.876\,\text{V}$ (5th)

▶ Falstad: `examples/samplehold/sh_capsize.txt` The too large capacitor is a low-pass filter rather than a hold stage: after every sample it has covered only $18\,\%$ of the remaining distance to the input voltage, $1 - e^{-20\,\text{ms}/102\,\text{ms}} = 0.178$. After five samples it shows $3\,\text{V}\,(1 - 0.822^5) = 1.88\,\text{V}$ instead of $3\,\text{V}$. The too small capacitor follows the input perfectly but loses $5\,\text{V/s}$ in the hold phase. A real leakage current is much smaller (pA to nA), so the lower limit for C_H is usually far away; we exaggerated it to make the effect visible.

Choosing C_H: the window

- upper limit from the acquisition, lower limit from the droop:

Hold capacitor

$$\frac{I_{\text{leak}}\, t_{\text{hold}}}{\Delta V_{\text{max}}} \;\le\; C_H \;\le\; \frac{t_{\text{track}}}{(R_S + R_{\text{on}})\,\ln 2^{N+1}}$$

- capstone values: $I_{\text{leak}} = 10\,\text{nA}$, $\Delta V_{\text{max}} = \frac{1}{10}\,\text{LSB} = 50\,\text{mV}$, $t_{\text{hold}} = 180\,\text{ms}$, $t_{\text{track}} = 20\,\text{ms}$, $R_S + R_{\text{on}} = 1.02\,\text{k}\Omega$
  - $36\,\text{nF} \le C_H \le 7.1\,\mu\text{F} \quad \Rightarrow \quad C_H = 1\,\mu\text{F}$ (geometric middle, film capacitor)
- choose the capacitor type: polypropylene or polystyrene film (low leakage, low dielectric absorption), C0G ceramic for small values; no electrolytic capacitors

Choosing a value in the geometric middle of the allowed window gives the same safety factor (here about 7) against both limits. Electrolytic capacitors are unsuitable: their leakage current is in the microampere range, and their capacitance is poorly defined. Dielectric absorption is a memory effect of the insulating material: after a fast change of voltage a part of the charge comes back slowly from the dielectric, and the held voltage creeps towards the previous value. Film capacitors with polypropylene or polystyrene dielectric show very little of it.

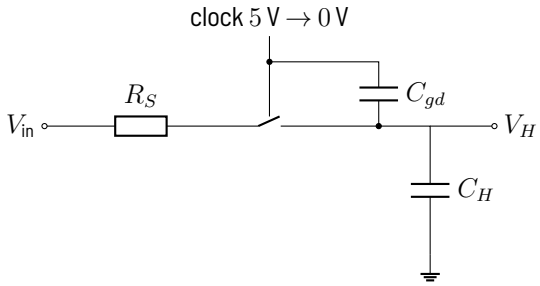
Design card: Sample and Hold

- structure: input follower $\to R_S$ + switch $\to C_H$ to ground $\to$ output follower
- clock: f_s, short duty cycle; track $t_{\text{track}} = D\,T_s$ (capstone: $5\,\text{Hz}$, $10\,\%$: $20\,\text{ms}$)
- charging: $\tau = (R_S + R_{\text{on}})\,C_H$, settled to $\frac{1}{2}\,\text{LSB}$ after $\tau\,\ln 2^{N+1}$ ($N = 3$: $2.77\,\tau$) $\le t_{\text{track}}$
- droop: $\Delta V = I_{\text{leak}}\, t_{\text{hold}}/C_H \le$ fraction of an LSB
- pedestal: $\Delta V = C_{gd}\,\Delta V_{\text{clk}}/(C_H + C_{gd})$; larger C_H helps
- capstone choice: $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$ film: $t_{\text{acq}} = 2.8\,\text{ms}$, droop $1.8\,\text{mV}$ at $10\,\text{nA}$

8.5 Errors of a Real Sample and Hold

Besides incomplete charging and droop, a real S&H stage shows a number of smaller errors. They matter in fast and high-resolution converters; in our slow 3-bit capstone they are negligible, but it is good to know where they come from and which design parameter influences them [6, 4].

Charge injection (pedestal)



- the gate of the switch is coupled to the hold node through C_{gd} (and the channel charge)
- when the switch opens, the clock step ΔV_{clk} is divided by C_{gd} and C_H:
  $\Delta V_{\text{ped}} = -\Delta V_{\text{clk}} \dfrac{C_{gd}}{C_{gd} + C_H} \approx -\dfrac{Q_{\text{inj}}}{C_H}$
- $C_{gd} = 10\,\text{pF}$, 5 V: $C_H = 1\,\text{nF}$: $-49.5\,\text{mV}$; $C_H = 100\,\text{nF}$: $-0.5\,\text{mV}$

▶ Falstad: `examples/samplehold/sh_pedestal.txt` The pedestal is a constant offset as long as the clock amplitude and C_{gd} are constant, so it can be calibrated. Its signal-dependent part (the channel charge depends on V_{GS}) cannot. Remedies are a larger C_H, small switches, a transmission gate (the charges of the n- and p-channel transistor partly cancel), and dummy switches driven with the inverted clock. In Falstad we model only the coupling capacitance; the labels `Vhold_1n` and `Vhold_100n` show 1.9505 V and 1.9995 V after the switch has opened, for an input of 2 V.

Feedthrough, aperture and jitter

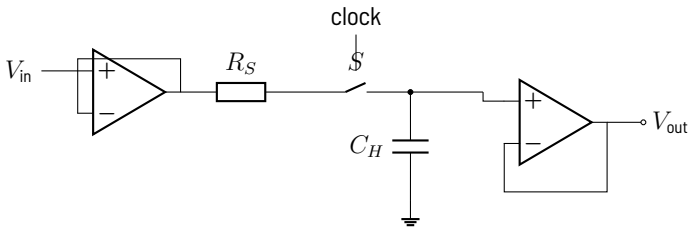
- **feedthrough**: the open switch has a capacitance C_{ds}; the input still reaches C_H: $\Delta V_H = \Delta V_{\text{in}} \frac{C_{\text{ds}}}{C_{\text{ds}}+C_H}$
- **aperture delay** t_a: the switch opens a little after the clock edge; constant, only shifts the sampling instant
- **aperture jitter** t_j: random variation of the sampling instant
  - error $\Delta V = \frac{dV}{dt}\, t_j$; sine: $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j$
  - requirement $\Delta V_{\text{max}} < \frac{1}{2}$ LSB: $t_j < \dfrac{1}{2\pi f\, 2^{N+1}} \cdot \dfrac{V_{\text{FS}}}{\hat{V}}$
  - example: 12 bit, full-scale sine of 1 MHz: $t_j < 19$ ps
- **dielectric absorption**: held voltage creeps back towards the previous value

For a full-scale sine $\hat{V} = V_{\text{FS}}/2$ the jitter limit is $t_j < 1/(2\pi f\, 2^N)$. Jitter is the reason why fast converters need clean, low-noise clock sources. For our 5 Hz system a jitter of a millisecond is harmless (exercise 8.6).

8.6 The Buffered Sample and Hold

The bare switch and capacitor only work with an ideal source and an ideal load. A real source has an internal resistance, for example the output resistance of a passive anti-aliasing filter, and a real load draws current, for example the input resistance of the converter or the reference ladder of a flash converter. The cure is a voltage follower on each side of the hold capacitor (chapter 6).

Follower – switch – capacitor – follower



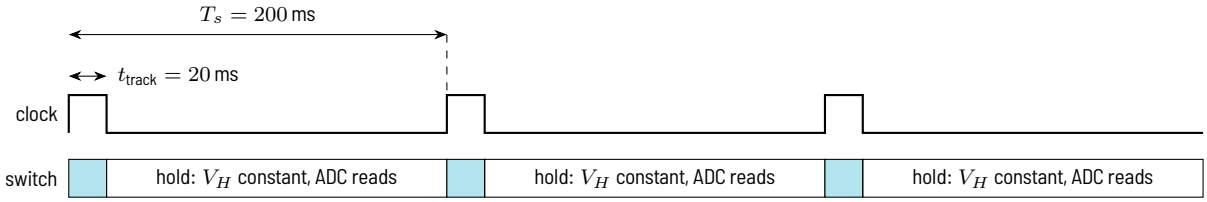
- **input buffer**: the charging current comes from the op-amp, not from the signal source
- **output buffer**: draws (almost) no current from C_H; only its input bias current adds to the leakage

Why both buffers are needed

- without buffers: source $R_i = 10\,\text{k}\Omega$, load $R_L = 100\,\text{k}\Omega$ (e.g. ADC input), $V_{\text{in}} = 3\,\text{V}$
  - track: $\tau = (R_i \parallel R_L)\, C_H = 9.1\,\text{ms}$, final value $3\,\text{V} \cdot \frac{100}{110} = 2.73\,\text{V}$
  - after $20\,\text{ms}$: $2.425\,\text{V}$
  - hold: discharge through R_L, $\tau = 100\,\text{ms}$: $0.401\,\text{V}$ after $180\,\text{ms}$
- with buffers: $3.000\,\text{V}$ during the whole period

▶ Falstad: `examples/samplehold/sh_buffered.txt` The unbuffered circuit fails in three ways at once: the source and the load form a voltage divider, the source resistance makes the charging slow, and the load discharges the capacitor in the hold phase. With buffers the source only sees the high input resistance of the first follower, and the capacitor only sees the high input resistance of the second follower. For the output buffer choose an op-amp with FET input (bias current in the pA range); its bias current is part of I_{leak}.

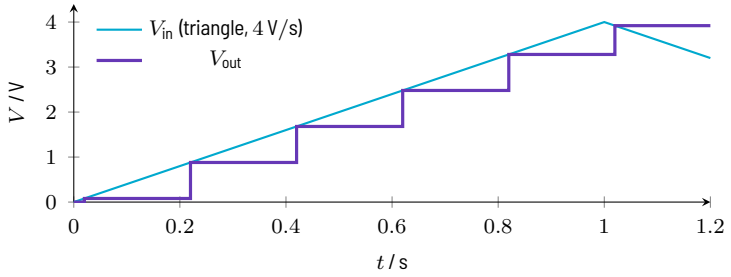
The sampling clock



- square wave with a short duty cycle $D = t_{\text{track}}/T_s$
- Falstad: rail with waveform *square*, $5\,\text{Hz}$, amplitude $2.5\,\text{V}$, offset $2.5\,\text{V}$, duty cycle 0.1
- in hardware: 555 timer or microcontroller output; 0/5 V logic level drives the switch
- the ADC result is read in the hold phase (e.g. in the middle or at the end)

A short track phase keeps the output constant for most of the period. The track phase must not be too short either: it must be longer than the acquisition time. With $D = 0.1$ at $5\,\text{Hz}$ we have $20\,\text{ms}$ for charging and $180\,\text{ms}$ of stable output.

Sampling a ramp



- buffered S&H, triangle $0\,\text{V} \ldots 4\,\text{V}$ at $0.5\,\text{Hz}$: steps of $4\,\text{V/s} \cdot 0.2\,\text{s} = 0.8\,\text{V}$
- Falstad: $0.08\,\text{V}$, $0.88\,\text{V}$, $1.68\,\text{V}$, $2.48\,\text{V}$, $3.28\,\text{V}$

▶ Falstad: `examples/samplehold/sh_staircase.txt` A step of $0.8\,\text{V}$ is larger than one LSB of the 3-bit converter ($0.5\,\text{V}$): consecutive samples skip codes. This is no error of the S&H stage. The input simply changes too fast for the sampling rate; such fast signals have to be removed by the anti-aliasing filter (see the sampling theorem in the next chapter).

Practical hints

- op-amps: rail-to-rail output for the full range $0\,\text{V} \ldots 4\,\text{V}$ on a single $5\,\text{V}$ supply, FET input for the output buffer
- guard ring around the hold node on the PCB: surface leakage
- integrated S&H amplifiers (LF398, AD783) contain both buffers and the switch; only C_H is external
- in most modern ADCs (SAR, sigma-delta) the S&H is built in: a switched capacitor array at the input
  - the driving op-amp must recharge this capacitor in the acquisition time: the same calculation as above

The LF398 is a classic monolithic S&H amplifier; its data sheet contains curves for acquisition time and droop rate as functions of C_H that are a good illustration of this chapter. Modern SAR converters in microcontrollers sample with an internal capacitor of a few picofarads; the source resistance must be low enough that this capacitor charges to $\frac{1}{2}$ LSB within the programmed sampling time, which is exactly the calculation $t_{\text{acq}} = \tau \ln 2^{N+1}$.

8.7 Exercises

Exercise 8.1: Choosing the hold capacitor for $f_s = 5\,\text{Hz}$

A buffered S&H stage samples at $f_s = 5\,\text{Hz}$ with a clock duty cycle of $10\,\%$. The charging path consists of $R_S = 1\,\text{k}\Omega$ and a switch with $R_{\text{on}} = 20\,\Omega$. The total leakage current at the hold node (switch, capacitor, buffer bias) is at most $I_{\text{leak}} = 10\,\text{nA}$. The stage drives a 3-bit converter with $V_{\text{FS}} = 4\,\text{V}$.

a) Calculate the track and hold times.

b) The capacitor must settle to $\frac{1}{2}$ LSB for a full-scale step within the track phase. Find the upper limit for C_H.

c) The droop during the hold phase must stay below $\frac{1}{10}$ LSB. Find the lower limit for C_H.

d) Choose C_H and calculate acquisition time and droop for your choice.

e) Check the charging curve in Falstad.

Solution 8.1

a) $T_s = 1/f_s = 200\,\text{ms}$, $t_{\text{track}} = 0.1 \cdot T_s = 20\,\text{ms}$, $t_{\text{hold}} = 180\,\text{ms}$.

b) LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$. Settling to $\frac{1}{2}$ LSB needs $t = \tau \ln 2^4 = 2.77\,\tau \le 20\,\text{ms}$, so $\tau \le 7.21\,\text{ms}$ and $C_H \le 7.21\,\text{ms}/1020\,\Omega = 7.1\,\mu\text{F}$.

c) $\Delta V_{\text{max}} = 50\,\text{mV}$: $C_H \ge I_{\text{leak}}\, t_{\text{hold}}/\Delta V_{\text{max}} = 10\,\text{nA} \cdot 0.18\,\text{s}/50\,\text{mV} = 36\,\text{nF}$.

d) The window is $36\,\text{nF}$ …$7.1\,\mu\text{F}$; the geometric middle is about $0.5\,\mu\text{F}$. We choose $C_H = 1\,\mu\text{F}$ (film capacitor). Then $\tau = 1.02\,\text{ms}$, $t_{\text{acq}} = 2.77\,\tau = 2.83\,\text{ms}$ (seven times faster than required) and $\Delta V_{\text{droop}} = 10\,\text{nA} \cdot 0.18\,\text{s}/1\,\mu\text{F} = 1.8\,\text{mV}$ (28 times smaller than allowed).

e) ▶ Falstad: `examples/samplehold/sh_acquisition.txt`
With a $4\,\text{V}$ input, `Vhold` shows $2.53\,\text{V}$ after $1.02\,\text{ms}$, $3.75\,\text{V}$ after $2.83\,\text{ms}$ and $4.00\,\text{V}$ at the end of the track phase. (Pause the simulation and use the scope, or reduce the simulation speed.)

Exercise 8.2: Droop and a too large capacitor

The S&H stage of exercise 8.1 has an (exaggerated) leakage current of $50\,\text{nA}$. The input is a constant $3\,\text{V}$.

a) Calculate the droop rate and the voltage at the end of the first hold phase for $C_H = 10\,\text{nF}$ and for $C_H = 1\,\mu\text{F}$.

b) With $C_H = 100\,\mu\text{F}$, which fraction of the remaining difference is charged in one track window? Calculate the held voltage after the first and after the fifth sample (leakage negligible).

c) Compare with Falstad.

Solution 8.2

a) $C_H = 10\,\text{nF}$: $dV/dt = 50\,\text{nA}/10\,\text{nF} = 5\,\text{V/s}$, droop in $180\,\text{ms}$: $0.9\,\text{V}$, the held voltage falls from $3.00\,\text{V}$ to $2.10\,\text{V}$. $C_H = 1\,\mu\text{F}$: $50\,\text{mV/s}$, droop $9\,\text{mV}$.

b) $\tau = 1.02\,\text{k}\Omega \cdot 100\,\mu\text{F} = 102\,\text{ms}$, fraction $a = 1 - e^{-20/102} = 0.178$. After n samples $V_n = 3\,\text{V}\,(1 - (1-a)^n)$: $V_1 = 0.534\,\text{V}$, $V_5 = 3\,\text{V}\,(1 - 0.822^5) = 1.876\,\text{V}$.

c) ▶ Falstad: `examples/samplehold/sh_capsize.txt`
`V_10n` drops from $3.00\,\text{V}$ to $2.10\,\text{V}$ in every hold phase, `V_1u` loses $9\,\text{mV}$ ($3.00\,\text{V} \to 2.99\,\text{V}$), `V_100u` climbs in steps: $0.534\,\text{V}$ after the first, $1.876\,\text{V}$ after the fifth sample.

Exercise 8.3: Unbuffered sample and hold

A switch and $C_H = 1\,\mu\text{F}$ are connected directly between a source ($V_{\text{in}} = 3\,\text{V}$, internal resistance $R_i = 10\,\text{k}\Omega$) and a load $R_L = 100\,\text{k}\Omega$. Clock: $5\,\text{Hz}$, $10\,\%$. Neglect R_{on}.

a) Draw the equivalent circuit for the track phase and calculate its final value and time constant.

b) Calculate V_H at the end of the first track phase and at the end of the first hold phase.

c) How does the buffered circuit change the result? Check in Falstad.

Solution 8.3

a) Thevenin equivalent seen by C_H: $V_{\text{th}} = 3\,\text{V} \cdot \frac{100}{110} = 2.727\,\text{V}$, $R_{\text{th}} = R_i \parallel R_L = 9.09\,\text{k}\Omega$, $\tau = 9.09\,\text{ms}$.

b) End of track: $V_H = 2.727\,\text{V}\,(1 - e^{-20/9.09}) = 2.425\,\text{V}$. Hold: C_H discharges through R_L with $\tau = 100\,\text{ms}$: $V_H = 2.425\,\text{V} \cdot e^{-1.8} = 0.401\,\text{V}$.

c) The input follower removes the divider and the slow charging, the output follower removes the discharge: $V_{\text{out}} = 3.000\,\text{V}$ throughout.

▶ Falstad: `examples/samplehold/sh_buffered.txt`

`Vout_unbuf` shows $2.425\,\text{V}$ and $0.401\,\text{V}$, `Vout_buf` $3.000\,\text{V}$.

Exercise 8.4: Pedestal error

The gate of the switch has a coupling capacitance $C_{gd} = 10\,\text{pF}$ to the hold node. The clock falls from $5\,\text{V}$ to $0\,\text{V}$ when the switch opens.

a) Calculate the pedestal for $C_H = 1\,\text{nF}$ and $C_H = 100\,\text{nF}$.

b) Which C_H is needed for a pedestal below $\frac{1}{2}$ LSB of a 12-bit converter with $V_{\text{FS}} = 4\,\text{V}$?

c) Is the pedestal a problem for the 3-bit capstone with $C_H = 1\,\mu\text{F}$?

Solution 8.4

a) $\Delta V = -5\,\text{V} \cdot \frac{10\,\text{pF}}{1.01\,\text{nF}} = -49.5\,\text{mV}$; with $100\,\text{nF}$: $-0.50\,\text{mV}$.

▶ Falstad: `examples/samplehold/sh_pedestal.txt`: $2\,\text{V}$ input, held values $1.9505\,\text{V}$ and $1.9995\,\text{V}$.

b) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{13} = 0.49\,\text{mV}$: $C_H \geq 5\,\text{V} \cdot 10\,\text{pF}/0.49\,\text{mV} = 0.1\,\mu\text{F}$.

c) $\Delta V = 50\,\mu\text{V}$, i.e. 10^{-4} of an LSB of $0.5\,\text{V}$: no problem.

Exercise 8.5: Ramp input and step size

A buffered S&H stage ($f_s = 5\,\text{Hz}$) samples a triangle that rises from $0\,\text{V}$ to $4\,\text{V}$ in $1\,\text{s}$.

a) Calculate the height of the steps of the output staircase.

b) The sampling instants are at the end of the track windows, $t_k = 20\,\text{ms} + k \cdot 200\,\text{ms}$. List the first five held values.

c) What is the maximum slope of the input if consecutive samples may differ by at most one LSB of the 3-bit converter? What does this mean for a sine with an amplitude of $2\,\text{V}$?

Solution 8.5

a) $4\,\text{V/s} \cdot 0.2\,\text{s} = 0.8\,\text{V}$.

b) $V(t_k) = 4\,\text{V/s} \cdot t_k$: $0.08\,\text{V}$, $0.88\,\text{V}$, $1.68\,\text{V}$, $2.48\,\text{V}$, $3.28\,\text{V}$.

▶ Falstad: `examples/samplehold/sh_staircase.txt`: `Vout` shows the same values ($4\,\text{mV}$ lower, because C_H lags by $\tau \cdot 4\,\text{V/s}$).

c) $0.5\,\text{V}/0.2\,\text{s} = 2.5\,\text{V/s}$. Sine: $2\pi f \cdot 2\,\text{V} \leq 2.5\,\text{V/s}$ gives $f \leq 0.2\,\text{Hz}$. Faster signals are still sampled correctly as long as $f < f_s/2$ (next chapter), but the codes of neighbouring samples jump.

Exercise 8.6: Aperture jitter

The $5\,\text{Hz}$ clock of the capstone comes from a microcontroller whose software has a timing uncertainty of $t_j = 1\,\text{ms}$. The input is a sine with $\hat{V} = 2\,\text{V}$ and $f = 2.5\,\text{Hz}$ (the highest frequency the anti-aliasing filter passes).

a) Calculate the maximum sampling error caused by the jitter.

b) Compare with $\frac{1}{2}$ LSB of the 3-bit converter.

c) Which jitter would a 16-bit converter tolerate for the same signal ($V_{\text{FS}} = 4\,\text{V}$)?

Solution 8.6

a) $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j = 2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V} \cdot 1\,\text{ms} = 31\,\text{mV}$.

b) $\frac{1}{2}\,\text{LSB} = 250\,\text{mV}$: the jitter error is eight times smaller, harmless.

c) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{17} = 30.5\,\mu\text{V}$: $t_j < 30.5\,\mu\text{V}/(2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V}) = 0.97\,\mu\text{s}$. A software-timed clock is no longer good enough; a hardware timer is needed.

Bibliography

[1] Paul Horowitz and Winfield Hill. *The Art of Electronics*. Cambridge University Press, 3rd edition, 2015. 8.3

[2] Ulrich Tietze, Christoph Schenk, and Eberhard Gamm. *Electronic Circuits: Handbook for Design and Application*. Springer, 2nd edition, 2008.

[3] Adel S. Sedra and Kenneth C. Smith. *Microelectronic Circuits*. Oxford University Press, 8th edition, 2020. 8.3

[4] Walt Jung, editor. *Op Amp Applications Handbook*. Analog Devices / Newnes, 2005. 8.5

[5] Op amps for everyone. Technical Report SLOD006B, Texas Instruments, 2002.

[6] Walt Kester, editor. *Data Conversion Handbook*. Analog Devices / Newnes, 2005. 8.1, 8.5

[7] Behzad Razavi. *Principles of Data Conversion System Design*. IEEE Press, 1995. 8.1

[8] Robert J. Widlar. New developments in IC voltage regulators. *IEEE Journal of Solid-State Circuits*, 6(1):2–7, 1971.

[9] A. Paul Brokaw. A simple three-terminal IC bandgap reference. *IEEE Journal of Solid-State Circuits*, 9(6):388–393, 1974.

[10] Claude E. Shannon. Communication in the presence of noise. *Proceedings of the IRE*, 37(1):10–21, 1949.

[11] Paul Falstad and Iain Sharp. CircuitJS1 – circuit simulator applet. `https://www.falstad.com/circuit/`, source code `https://github.com/pfalstad/circuitjs1`. accessed October 2026.

[12] Franco Maloberti. *Data Converters*. Springer, 2007.

[13] Robert H. Walden. Analog-to-digital converter survey and analysis. *IEEE Journal on Selected Areas in Communications*, 17(4):539–550, 1999.

[14] IEC. IEC 60751: Industrial platinum resistance thermometers and platinum temperature sensors. International Electrotechnical Commission, 2022.

[15] R. P. Sallen and E. L. Key. A practical method of designing RC active filters. *IRE Transactions on Circuit Theory*, 2(1):74–85, 1955.

[16] Charles Kitchin and Lew Counts. A designer's guide to instrumentation amplifiers. Technical report, Analog Devices, 2006.